

Linear System Theory

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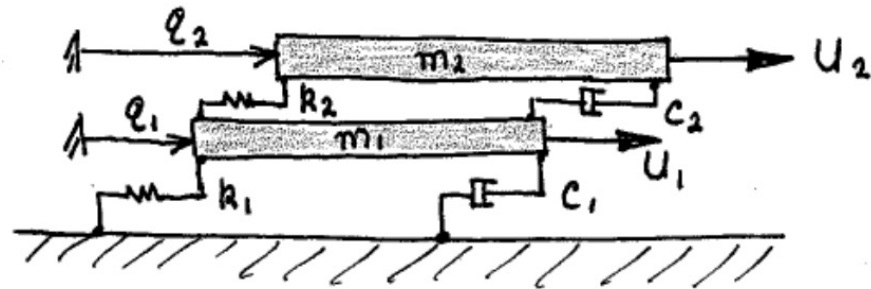
Outline

- Introduction
- State space representation
- Linear time-invariant (LTI) system
- Transfer function



Linear system – Simple structure (HW2)

$$\begin{aligned} m_1 \ddot{q}_1 + (c_1 + c_2) \dot{q}_1 + (k_1 + k_2) q_1 - c_2 \dot{q}_2 - k_2 q_2 &= u_1 \\ m_2 \ddot{q}_2 - c_2 \dot{q}_1 - k_2 q_1 + c_2 \dot{q}_2 + k_2 q_2 &= u_2 \end{aligned}$$



Discretization – Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n$$
$$= f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!} (x-x_0)^2 + \dots$$

$$\text{Let } x(t_0), \dot{x}(t_0)$$

$$\Rightarrow x(t_0 + \Delta t) \approx x(t_0) + \dot{x}(t_0) \Delta t$$

If we have $\ddot{x}(t_0), \dddot{x}(t_0), \dots$

$$x(t_0 + \Delta t) = x(t_0) + \dot{x}(t_0) \Delta t + \frac{1}{2!} \ddot{x}(t_0) \Delta t^2 + \frac{1}{3!} \dddot{x}(t_0) \Delta t^3 + \dots$$





Discretization

$$\dot{x} = f(x(t), u(t))$$

Euler discretization = 1st order Taylor

$$\dot{x}(t) = u(t) \quad kT \leq t < (k+1)T$$

$$\begin{aligned} \underline{x((k+1)T)} &= x(kT) + \int_{kT}^{(k+1)T} \dot{x}(\tau) d\tau \\ &= x(kT) + \int_{kT}^{(k+1)T} u(\tau) d\tau \\ &= x(kT) + \int_{kT}^{(k+1)T} u[k] d\tau \\ &= \underline{x[k]} + u[k] \cdot T \end{aligned}$$

" $x[k+1]$

$x[k]$



$$m\ddot{q} = u \quad \dot{x}_1 = \underline{\underline{x_2}} \quad x_2(t) \quad x_1[k+1] = x_1[k] + T[x_2[k] + \frac{T^2}{2m} u[k]]$$

Discretization $\ddot{x}_2 = \frac{1}{m} u \rightarrow \text{EOL}$

$$kT \leq t < (k+1)T$$

$$x_2[k+1] = x_2[k] + \frac{T}{m} u[k]$$

$$x_2(t) = x_2(kT) + \int_{kT}^t \ddot{x}_2(\tau) d\tau$$

$$\uparrow$$

$$= x_2(kT) + \int_{kT}^t \frac{1}{m} u[k] d\tau$$

$$(k+1)T$$

$$= \underline{\underline{x_2[k]}} + \frac{(t-kT)}{m} u[k]$$

$$x_1(t) = x_1(kT) + \int_{kT}^t \dot{x}_1(\tau) d\tau$$

$$\uparrow$$

$$= x_1(kT) + \int_{kT}^t x_2[k] + \frac{(t-kT)}{m} u[k] d\tau$$

$$= x_1[k] + x_2[k] + \frac{(t-kT)^2}{2m} u[k]$$



Equilibrium State

- Consider a system $\dot{x} = f(x)$
- An equilibrium solution is the simplest type of solution, it is **constant for all time**, that is, it satisfies

$$x(t) = x^e$$

for some fixed state vector. The state is called **equilibrium state**.

- Since an equilibrium solution must satisfy the above differential equation, all equilibrium states must satisfy the equilibrium condition:

$$f(x^e) = 0 \quad \Rightarrow \text{不再變化}$$



Controlled equilibrium states

- Consider a system $\dot{x} = f(x, u)$
- A state x^e is a *controlled equilibrium* state of the system, if there is a constant input u^e such that

$$f(x^e, u^e) = 0$$



Equilibrium state (Discrete System)

- Consider a discrete-time system described by
$$x(k + 1) = f(x(k))$$
- All equilibrium states must satisfy the equilibrium condition:
$$x(k) = x^e = f(x^e)$$
- In continuous time, equilibrium means **zero rate of change**, while in discrete time, it means **the next state equals the current state**.

$$x(k+1) = x^e$$



Time-invariant systems $\Leftrightarrow$ Time-varying sys

- **Time-invariant:** If the same input applied at different times produces identical outputs (shifted in time).
- **State and Input Relationship:**

Input at time $t_0 \rightarrow$ Output $y(t)$

Input shifted to $t_0 + T \rightarrow$ Identical output shifted by T



Linear Time-invariant system (LTI)

A "finite-dimensional" continuous-time, LTI

input-output sys

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx + Du\end{aligned}$$

$$\begin{aligned}\dot{x} &= \underline{A(t)}x(t) + \underline{B(t)}u(t) \\ y &= \underline{C(t)}x(t) + \underline{D(t)}u(t)\end{aligned}$$

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

$$A(t) = \begin{bmatrix} 2t & t^2 \\ 0 & 1 \end{bmatrix}$$

discrete-time LTI

$$\begin{aligned}x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k)\end{aligned}$$



Linearization about an equilibrium solution

Consider a m -vector-valued func. $f(x): \mathbb{R}^n \rightarrow \mathbb{R}^m$ $x_1, x_2, \dots, x_n$

The derivative of $f(\cdot)$ (Jacobian of f) at x^* $f_1, f_2, \dots, f_m$

$$Df(x^*) = \frac{\partial f}{\partial x}(x^*) := \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(x^*) & \frac{\partial f_1}{\partial x_2}(x^*) & \dots & \frac{\partial f_1}{\partial x_n}(x^*) \\ \frac{\partial f_2}{\partial x_1}(x^*) & & & \\ \vdots & & & \\ \frac{\partial f_m}{\partial x_1}(x^*) & & & \frac{\partial f_m}{\partial x_n}(x^*) \end{bmatrix}$$

$$Df(x^*)_{ij} = \frac{\partial f_i}{\partial x_j}(x^*)$$



Linearization about an equilibrium solution

$$Df(x^*)_{ij} = \frac{\partial f_i}{\partial x_j}(x^*)$$

the i - j th element of Df is the partial derivative of f_i with respect to x_j

* All linearization is based on the following fundamental property of the derivative.
 x, x^* are any two vectors, x is "close" to " x^* "

$$f(x) \approx \boxed{f(x^*)} + \underline{\underline{\frac{\partial f}{\partial x}(x^*) (x - x^*)}}$$



Linearization about an equilibrium solution

$$f(x) = \begin{bmatrix} x_1 x_2 \\ \cos(x_1) \\ e^{x_2} \end{bmatrix} \quad \begin{matrix} f_1 \\ f_2 \\ f_3 \end{matrix} \quad \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

x_1, x_2 f_1, f_2, f_3

$$Df(x) = \frac{\partial f_i}{\partial x_j} (x) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \vdots & \vdots \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} \end{bmatrix} \quad \begin{matrix} 3 \\ 3 \times 2 \end{matrix} = \begin{bmatrix} x_2 & x_1 \\ -\sin(x_1) & 0 \\ 0 & e^{x_2} \end{bmatrix}$$

$x^* = x$

$$D(f+g) = Df + Dg$$

$$D(\alpha f) = \alpha Df$$

constant func. $f(x) = c$ $Df(x) = 0$

Linear func $f(x) = Ax$ $Df(x) = A$



Linearization about an equilibrium solution

$$\begin{cases} \dot{x} = F(x, u) \\ y = H(x, u) \end{cases} \quad \begin{array}{l} x^e, u^e \leftarrow \text{equilibrium state,} \\ \text{const. input} \end{array} \Rightarrow F(x^e, u^e) = 0$$

$y = H(x^e, u^e)$

x^e, u^e, y^e = trim condition

$\delta x, \delta u, \delta y$ = perturbed State/Input/output

$$\begin{aligned} \delta x &= x - x^e, \quad \delta u = u - u^e, \quad \delta y = y - y^e \\ \hookrightarrow x &= x^e + \delta x, \quad u = u^e + \delta u, \quad y = y^e + \delta y \end{aligned} \quad \left. \begin{array}{l} \dot{\bar{x}} = \dot{x}^e + \delta \dot{x} \\ = F(x^e + \delta x, u^e + \delta u) \\ y = y^e + \delta y \\ = H(x^e + \delta x, u^e + \delta u) \end{array} \right\}$$



$$\delta y = H(x^e + \delta x, u^e + \delta u) - H(x^e, u^e)$$

$$\delta \dot{x} = \underline{F(x^e + \delta x, u^e + \delta u)}$$

$$\delta y = \underline{H(x^e + \delta x, u^e + \delta u)} - \cancel{H(x^e, u^e)} \quad \delta x, \delta u, \delta y \text{ are "small"}$$

$$F(x^e + \delta x, u^e + \delta u) \approx \underline{F(x^e, u^e)} + \frac{\partial F}{\partial x}(x^e, u^e) \delta x + \frac{\partial F}{\partial u}(x^e, u^e) \delta u$$

$$\delta \dot{x} \approx \frac{\partial F}{\partial x}(x^e, u^e) \delta x + \frac{\partial F}{\partial u}(x^e, u^e) \delta u$$

$$H(x^e + \delta x, u^e + \delta u) \approx \cancel{H(x^e, u^e)} + \frac{\partial H}{\partial x}(x^e, u^e) \delta x + \frac{\partial H}{\partial u}(x^e, u^e) \delta u$$

$$\delta y \approx \frac{\partial H}{\partial x}(x^e, u^e) \delta x + \frac{\partial H}{\partial u}(x^e, u^e) \delta u$$



Linearization of Discrete-Time Systems

$$x(k+1) = F(x(k), u(k))$$

$$y(k) = H(x(k), u(k))$$

x^e, u^e, y^e trim Cond.

$\delta x, \delta u, \delta y$ perturbed state / input / output

$$F(x^e, u^e) = x^e$$

$$H(x^e, u^e) = y^e$$

$$x(k+1) = \cancel{x^e} + \boxed{\delta x(k)}$$

$$= F(x^e + \delta x(k), u^e + \delta u(k))$$

$$= \underbrace{F(x^e, u^e)}_{x^e} + \frac{\partial F}{\partial x}(x^e, u^e) \delta x(k) + \frac{\partial F}{\partial u}(x^e, u^e) \delta u(k)$$

$$\delta x(k+1) = \frac{\partial F}{\partial x}(x^e, u^e) \delta x(k) + \frac{\partial F}{\partial u}(x^e, u^e) \delta u(k)$$

$$\delta y(k) = \frac{\partial H}{\partial x}(x^e, u^e) \delta x(k) + \frac{\partial H}{\partial u}(x^e, u^e) \delta u(k)$$



Example – simple pendulum

$$F(x): \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

Example 6 (Controlled pendulum)

$$D = \begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -\sin(x_1) \end{cases} \Rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -\sin(x_1) \end{bmatrix}$$

Jacobian $A = \frac{\partial \dot{F}}{\partial x}(x) = \begin{bmatrix} 0 & 1 \\ -\cos(x_1) & 0 \end{bmatrix}_{2 \times 2}$

Equilibrium state: $x^c = \begin{bmatrix} 0 \\ \pi \end{bmatrix} = k\pi$ ① $x^c = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, A(x^c) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
 ② $x^c = \begin{bmatrix} \pi \\ 0 \end{bmatrix}, A(x^c) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$



Example – Discrete Time System

Example 22 Simple scalar nonlinear system

$$f(x) = x(k+1) = x(k)^2 \quad \therefore \mathbb{R}^1 \rightarrow \mathbb{R}^1$$

Jacobian $\frac{\partial f}{\partial x}(x) = 2x$

Equilibrium State $x^e = x^e \quad x^e = 0, 1$

For $x^e = 0$ $\delta x(k+1) = \frac{\partial f}{\partial x}(x^e) \delta x(k)$
 $= 0$

For $x^e = 1$ $\delta x(k+1) = 2 \delta x(k)$



$$F(q, \dot{q}, \ddot{q}, u) = \ddot{q} + s'(q)u - u = 0$$

Implicit linearization $\hookrightarrow \delta \ddot{q} + \cos(q^e) \delta q - \delta u = 0 \quad q^e = 0$

$$\delta \ddot{q} + \delta q - \delta u = 0$$

- When we have a higher order ODE, it is usually easier to first linearize and then obtain a state space description than vice-versa.

single equation

$$F(q, \dot{q}, \ddot{q}, \dots, q^{(n)}, u) = 0$$

$$q(t) \equiv q^e, \quad u(t) \equiv u^e$$

$$\frac{\partial F}{\partial q}(\ast) \delta q + \frac{\partial F}{\partial \dot{q}}(\ast) \delta \dot{q} + \dots + \frac{\partial F}{\partial q^{(n)}}(\ast) \delta q^{(n)} + \frac{\partial F}{\partial u}(\ast) \delta u = 0$$

$$(\ast) = (q^e, 0, 0, \dots, 0, u^e)$$

$$\hookrightarrow \dot{q}^e = 0, \ddot{q}^e = 0, \dots, q^{(n)e} = 0$$



Example – Orbit mechanics

$$\omega^2 = \mu / r^3$$

$$\begin{cases} \ddot{r} - r\omega^2 + \mu/r^2 = 0 \\ r\dot{\omega} + 2\dot{r}\omega = 0 \end{cases}$$

scalar $\vec{r}^e, \vec{\omega}^e \rightarrow \ddot{\vec{r}}^e, \ddot{\vec{\omega}}^e \dots$

$$\begin{cases} \delta \ddot{r} - \omega^2 \delta r - 2\omega^e r^e \delta \omega - \left(2\mu / (r^3) \right) \delta r = 0 \\ \cancel{\dot{\omega}^e} \delta r + r^e \delta \dot{\omega} + 2\omega^e \delta \dot{r} + \cancel{2\dot{r}^e} \delta \omega = 0 \\ \delta \dot{r} - 3\omega^2 \delta r - 2\omega^e r^e \delta \omega = 0 \\ r^e \delta \dot{\omega} + 2\omega^e \delta \dot{r} = 0 \end{cases}$$

$$\begin{aligned} x_1 &= \delta r \\ x_2 &= \delta \omega \\ x_3 &= \delta \dot{r} \end{aligned}$$



Transfer Function

- For linear time-invariant systems, transfer functions provide another way of describing input-output systems
 - State space description: Time-domain
 - Transfer function: Frequency-domain
- To discuss transfer functions of continuous-time systems, we need the *Laplace transform*

$$\hat{f}(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} \underline{f(t)} dt$$

scalar, vector,
matrix



Laplace transform

- Note that the Laplace transform is a linear operator

$$\mathcal{L}(f+g) = \mathcal{L}(f) + \mathcal{L}(g)$$

- Also, if M is any constant matrix of appropriate dimensions

$$\mathcal{L}(Mf) = M \mathcal{L}(f)$$

- We will need the following very useful relationship:

$$\mathcal{L}(\dot{f})(s) = s \hat{f}(s) - f(0)$$

$$* f(t) = e^{at}, \quad a \text{ is any scalar}, \quad f(0) = 1$$

$$\dot{f} = af \rightarrow s \hat{f}(s) - 1 = a \hat{f}(s)$$

$$\hat{f}(s) = \frac{1}{s-a}$$



1931

成功大學
National Cheng Kung University

Transfer Function

Consider now the LTI system

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) + Du(t) \end{cases}$$

A, B, C, D are matrices (can be scalar)

along with the initial condition

$$x(0) = x_0.$$

Taking the Laplace transform

$$\begin{aligned} s\hat{x}(s) - x_0 &= A\hat{x}(s) + B\hat{u}(s) \\ \hat{y}(s) &= C\hat{x}(s) + D\hat{u}(s) \end{aligned} \quad \left\{ \begin{aligned} (sI - A)\hat{x}(s) &= x_0 + B\hat{u}(s) \\ \hat{y}(s) &= C\hat{x}(s) + D\hat{u}(s) \end{aligned} \right.$$

The **Transfer function (matrix)** is defined by

$$\hat{G}(s) = C(sI - A)^{-1}B + D$$

$$\hat{y}(s) = C(sI - A)^{-1}x_0 + \hat{G}(s)\hat{u}(s)$$



Zero initial conditions response

$$\hat{y}(s) = C(sI - A)^{-1}x_0 + \hat{G}(s)\hat{u}(s)$$

- Suppose that $x(0) = 0$. Then the relationship between the Laplace transform of the input and the output is given by a simple relationship

$$\hat{y} = \hat{G} \hat{u}$$

- For a SISO system, for $u \neq 0$,

$$\hat{G} = \frac{\hat{y}}{\hat{u}}$$



Example 25

$$\begin{aligned}\dot{x}(t) &= -x(t) + u(t) \\ y(t) &= x(t)\end{aligned}$$

$$A = -1, B = 1, C = 1, D = 0$$

$$\begin{aligned}\hat{G}(s) &= C(sI - A)^{-1}B + D = [1 \cdot (s+1)^{-1}] \cdot 1 + 0 \\ &= \frac{1}{s+1}\end{aligned}$$

Example 26 Unattached mass

$$\frac{ss^2 + f}{\updownarrow}$$

$$+ f_2 s s$$

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= u \\ y &= x_1\end{aligned}$$

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$C = [1 \ 0] \quad D = 0$$

$$\hat{G}(s) = [1 \ 0] \begin{bmatrix} s & -1 \\ 0 & s \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + D = \frac{1}{s^2}$$



$$C_0 s^0 + C_1 s + C_2 s^2 + \dots + C_n s^n$$

Rational transfer functions

分子 & 分母 都是 polynomials

- A scalar valued function of a complex variable is rational if it can be expressed as the ratio of two polynomials

- **Proper**: the order of the numerator polynomial
is less than the order of the denominator polynomial

- **Strictly Proper**:

the order of the numerator polynomial
is less than the order of the denominator polynomial



polynomial : $C_0 s^0 + C_1 s^1 + C_2 s^2 + \dots + C_n s^n$

Transfer function examples

Example 28

$$\hat{G}(s) = \frac{s^1}{s^2+1}$$

Rational

✓

$1 < 2$, strictly proper

$$\hat{G}(s) = \frac{s^2}{s^2+1}$$

✓

proper

$$\hat{G}(s) = \frac{s^3}{s^2+1}$$

✓

$3 > 2$ not proper

$$\hat{G}(s) = \frac{e^{-s}}{s^2+1}$$

✗

* The TF of a finite-dimensional LTI is always rational & proper



General LTI input output system

- $\hat{G}(s) = C(sI - A)^{-1}B + D = \frac{1}{\Delta(s)} N(s)$
 - $\Delta(s) = \det(sI - A)$: characteristic polynomial of A
 - $\deg(N(s)) \leq \deg(\Delta(s))$ proper
- Note that: state space to TF is unique, but TF to state space might not be unique

$$\frac{N(s)}{\Delta(s)} = \frac{2N(s)}{2\Delta(s)} = \frac{sN(s)}{\Delta(s)} = \text{TF}$$

$\downarrow$ $\downarrow$ $\downarrow$
 SS 1 SS 2 SS 3



Pole and Zeros

$$\frac{(sI - A)^{-1}}{\det(sI - A)}$$

- Poles: $\lim_{s \rightarrow \infty} \hat{G}_{ij}(s) = \infty \Rightarrow \text{denominator of } \hat{G}_{ij}(s) = 0$
 - Determine the system's natural dynamics & stability
- Zeros: $\hat{G}(u) = 0$
 - Here, we only define zeroes for scalar-input scalar-output transfer functions.
 - The definition of zeroes for MIMO systems is much more complicated
 - Determine how the input is shaped or canceled in the output



Discrete time transfer function

Consider a discrete time LTI system

$$\begin{aligned}x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k)\end{aligned}$$

- The transfer function matrix is defined by

$$\hat{G}(z) = C(zI - A)^{-1}B + D$$

$z \in \mathbb{C}$. This is the z -transform of the impulse response matrix g .

